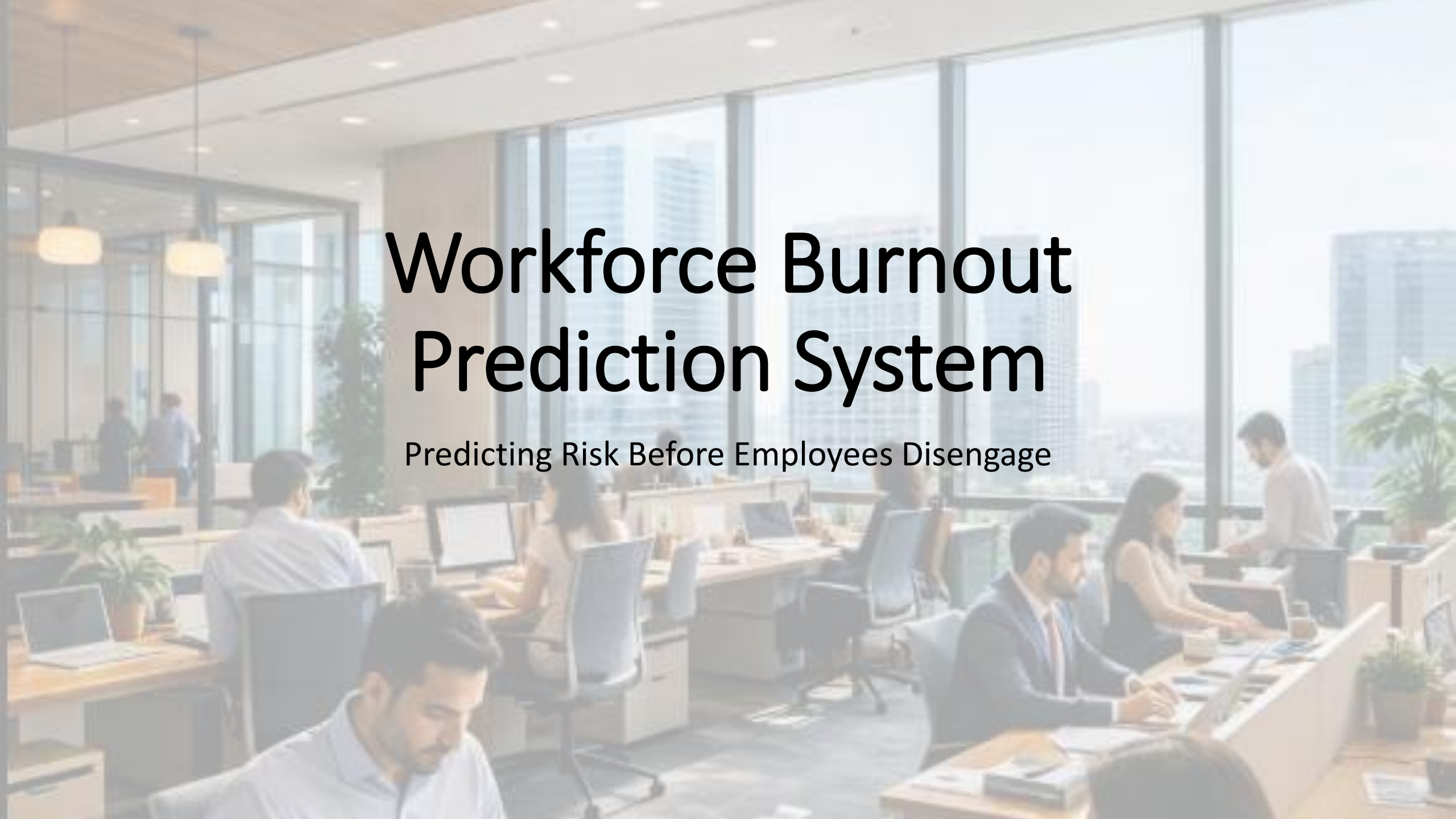


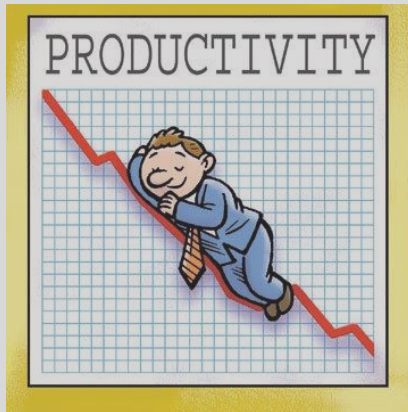


Workforce Burnout Prediction System

Predicting Risk Before Employees Disengage

Business understanding

Employee Burnout Is rising, organizations across industries are facing increasing challenges related to



Why this matters

Burnout affects:

- ❖ Business performance
- ❖ Retention costs
- ❖ Team morale
- ❖ Employee health outcomes
- ❖ Long-term organizational stability

Objectives

1

Identify Burnout

2

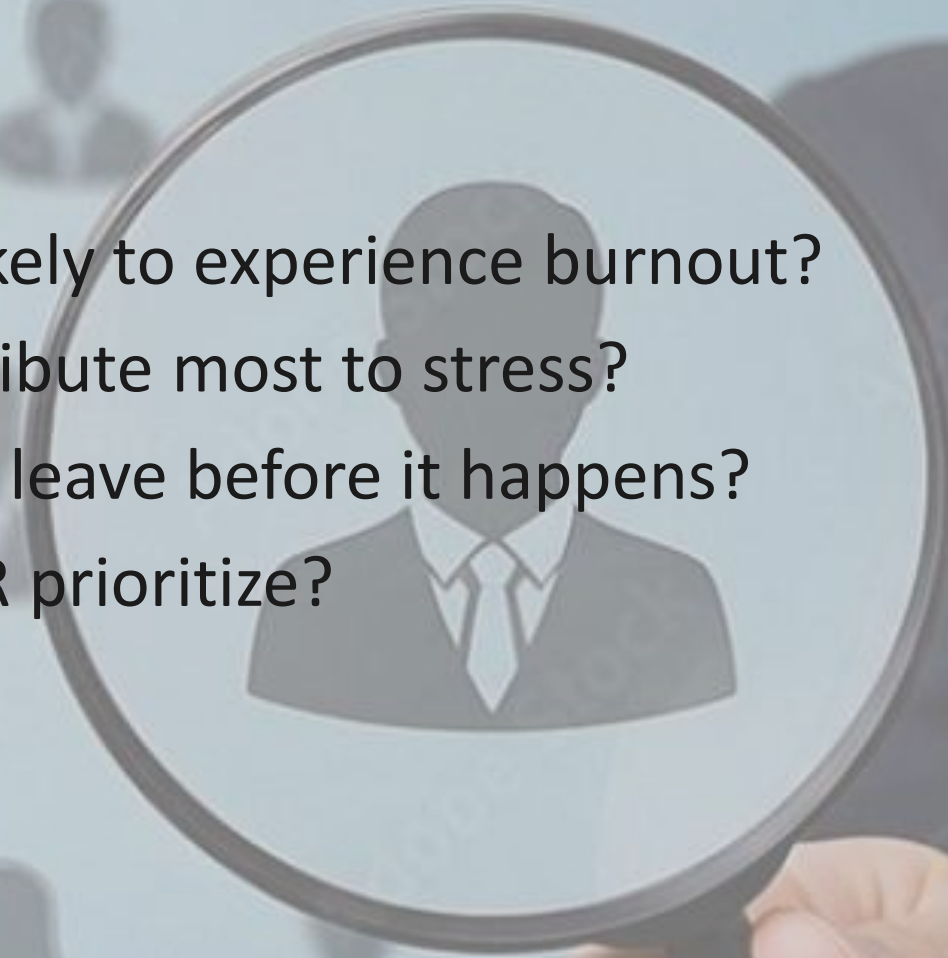
Forecast Sick Days

3

Recommend HR Actions

Key questions

- Which employees are most likely to experience burnout?
- What workplace factors contribute most to stress?
- Can we predict increased sick leave before it happens?
- What interventions should HR prioritize?



Dataset overview

- 1,470 Employees
- 35 Variables
- HR Analytics Dataset

Understanding Burnout Risk

A burnout risk framework using workplace indicators such as:

- Overtime frequency
- Work-life balance
- Job satisfaction
- Environment satisfaction
- Income level
- Travel demands

➤ Employees were grouped into:

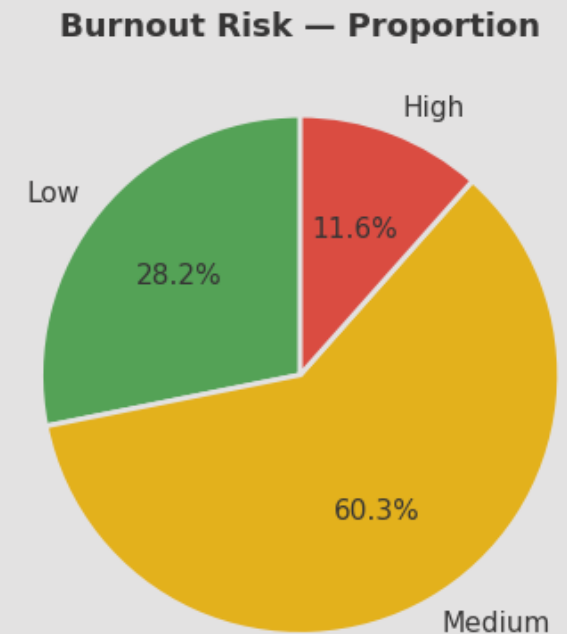
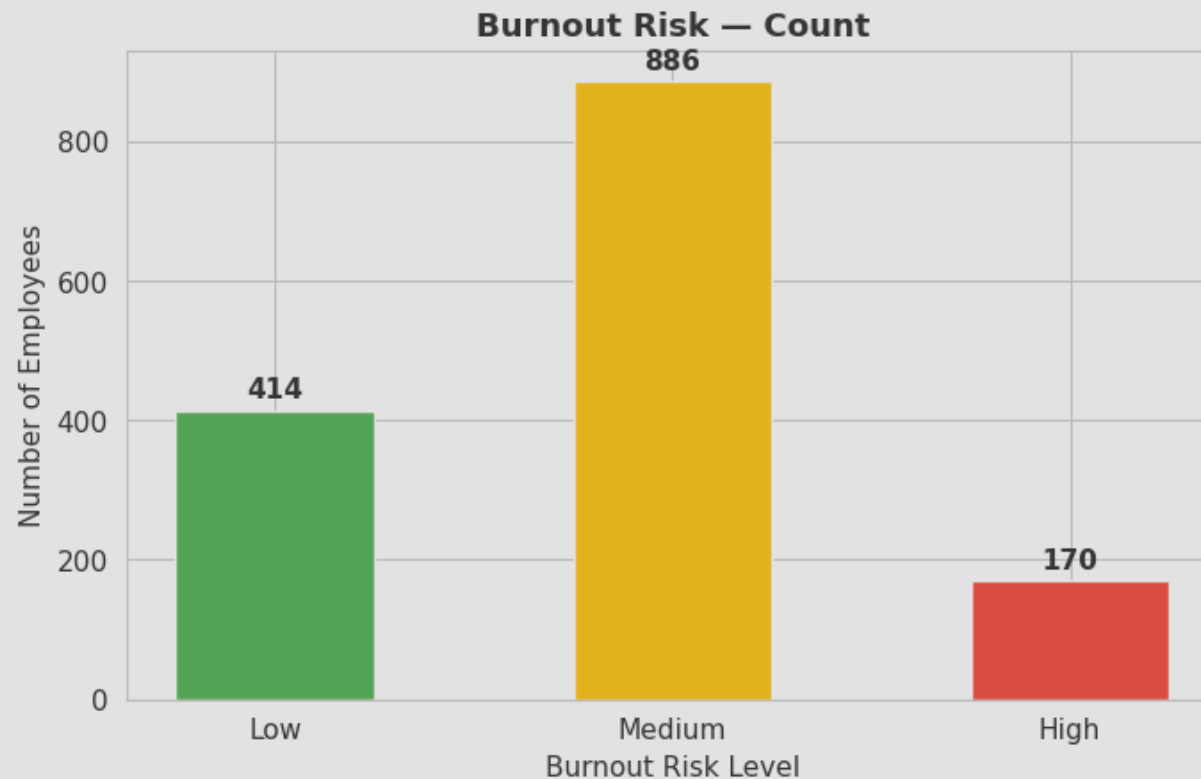
- Low Risk - Healthy engagement and stable work conditions
- Medium Risk - Early signs of stress or disengagement
- High Risk - Strong indicators of burnout and potential attrition

What was found

A significant portion of the workforce may already be experiencing stress-related challenges.

This creates an opportunity for early intervention before productivity and retention decline.

BurnoutRisk Class Distribution

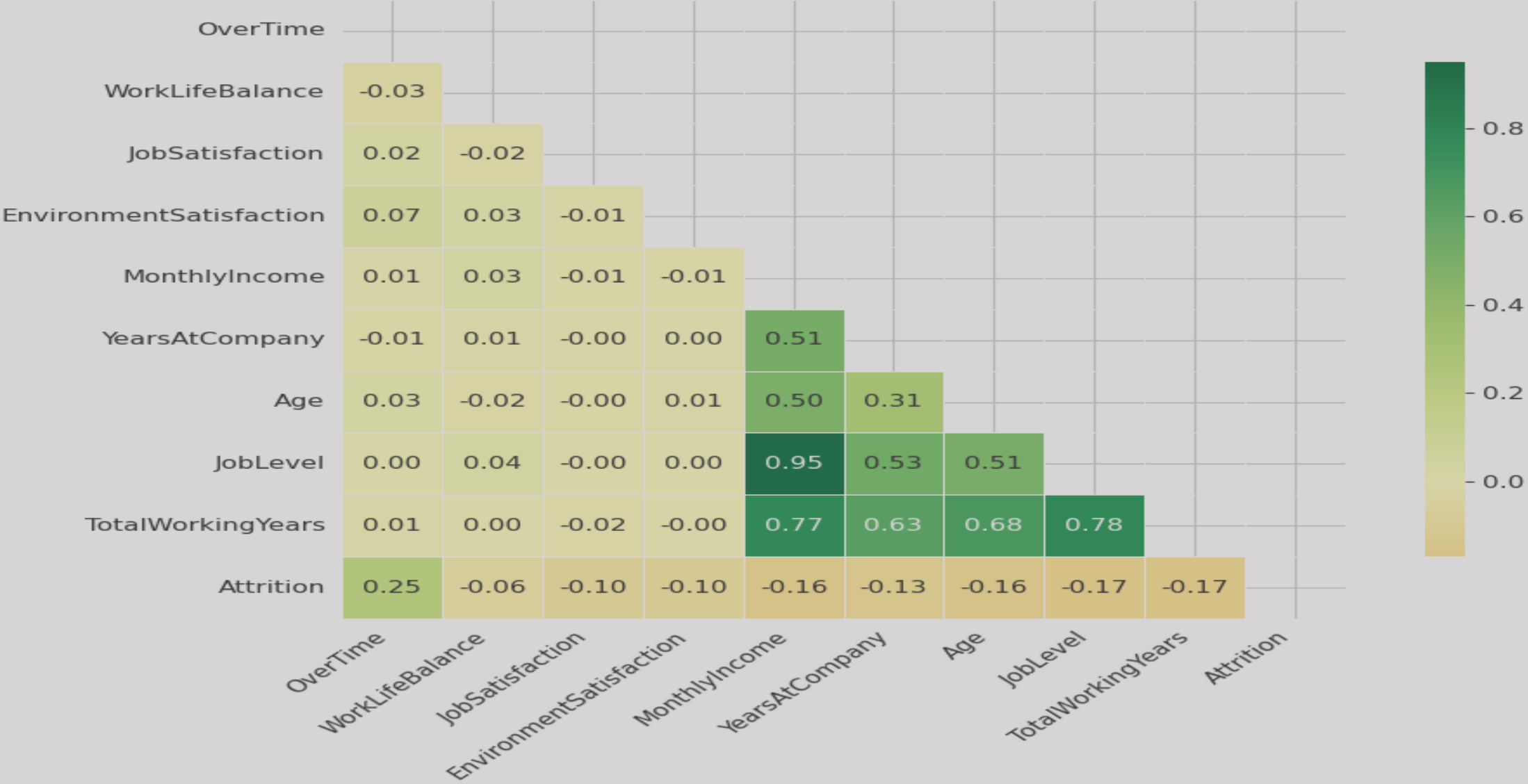


Major Drivers of Burnout

The analysis identified several major contributors:

- Overtime - Employees working overtime were significantly more likely to experience burnout.
- Poor Work-Life Balance - Employees reporting poor balance showed the highest concentration of burnout risk.
- Low Job Satisfaction- Lower satisfaction strongly aligned with higher stress levels.
- Lower Income Levels - Higher burnout risk was more concentrated among lower-income employees

Correlation Heatmap — Key HR Features



Burnout & Employee Attrition

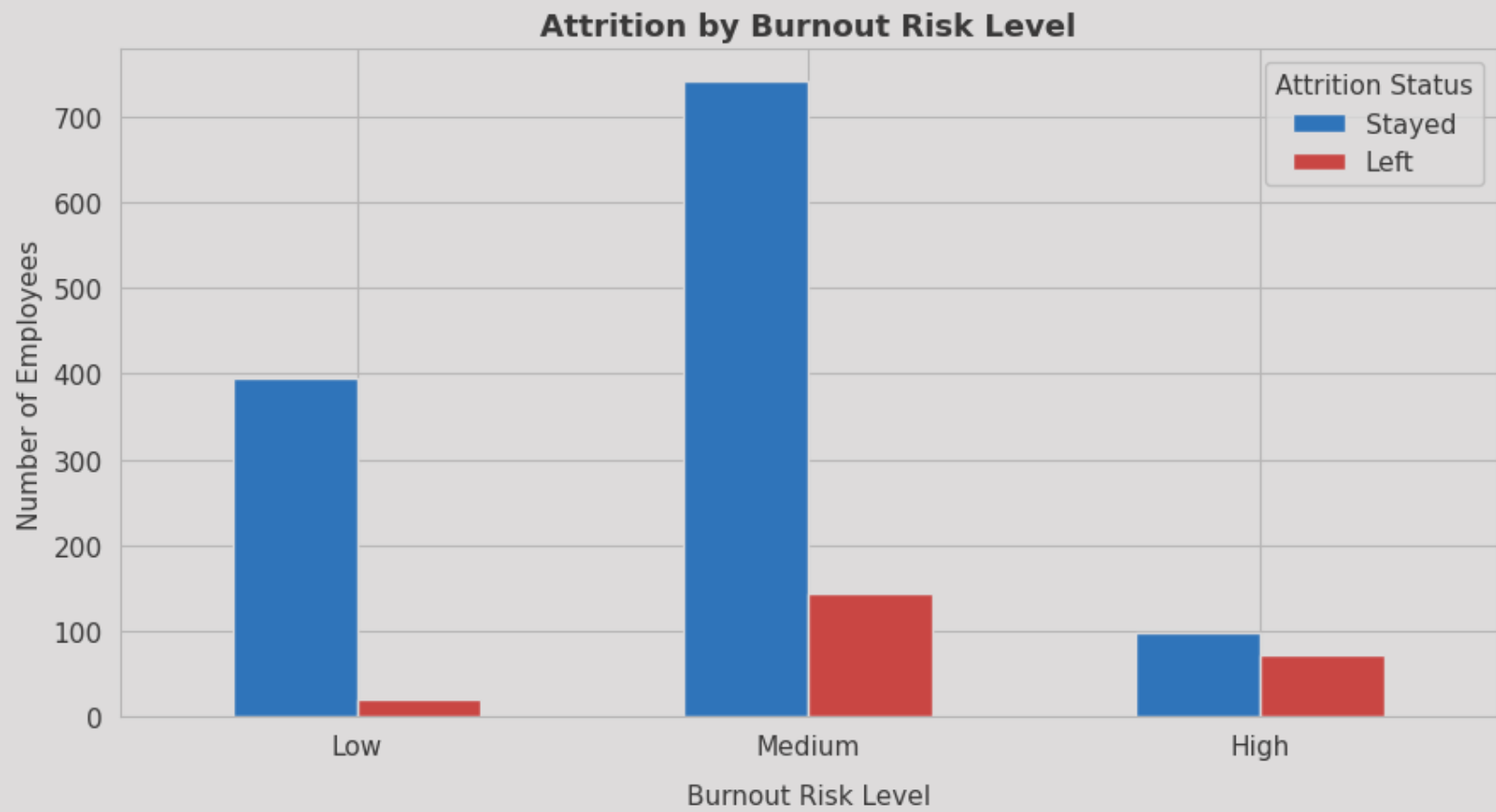
- Employees classified as high burnout risk were far more likely to leave the organization.

Key Insight

- Burnout is not only a wellness issue — it is also a retention and operational risk.

Potential organizational impacts include:

- Increased hiring costs
- Loss of experienced staff
- Reduced team continuity
- Lower productivity



Predictive Modeling Turning Data Into Early Warning Signals

We developed predictive models capable of:

- Identifying burnout patterns
- Forecasting future sick-day trends
- Supporting HR decision-making

➤ Why This Matters

- Instead of waiting until employees resign or disengage, organizations can identify warning signs earlier and respond proactively.

Sick-Day Forecasting Predicting Future Absenteeism

- The system estimated potential sick-day patterns based on workplace conditions and burnout indicators.
- Main Insight Employees with higher burnout risk were more likely to experience increased absenteeism.
- This allows organizations to:
 - Anticipate workforce disruptions
 - Improve staffing plans
 - Reduce productivity losses
 - Support employee wellbeing earlier

HR Recommendation Engine From Prediction to Action

- The project included a recommendation system that suggested interventions based on employee risk levels.

Examples of Recommendations

- Immediate HR follow-up
- Mental wellness support
- Workload review
- Overtime reduction
- Moderate Risk Manager check-ins
- Flexible scheduling

Priority groups identification

Workforce Segmentation

The system categorized employees into intervention priority groups:

- Critical
- High
- Moderate

Key Finding

- A meaningful number of employees required urgent attention, suggesting that burnout risks may already be affecting organizational performance.

Business Value

- Potential Organizational Benefits
 - Improved Employee Retention
 - Identify struggling employees before they leave.
 - Better Workforce Planning
 - Forecast absenteeism and staffing needs.
 - Reduced Burnout Costs
 - Lower productivity losses and turnover expenses.
- Data-Driven HR Decisions move beyond intuition and use measurable workforce insights.
- Stronger Employee Wellbeing Create a healthier and more supportive work culture.

Recommendations

1. Monitor Overtime Closely

Excessive overtime consistently emerged as a major risk factor.

2. Strengthen Work-Life Balance Policies

Flexible scheduling and workload management can reduce burnout.

3. Increase Employee Support Programs

Expand mental wellness and employee assistance initiatives.

4. Use Predictive Analytics in HR

Adopt early-warning systems to support proactive intervention.

5. Regularly Measure Employee Satisfaction

Continuous feedback helps identify emerging risks early

Burnout Is Predictable — And Preventable

This project demonstrates that workforce data can help organizations:

- Detect burnout earlier
- Anticipate absenteeism
- Improve employee wellbeing
- Reduce turnover risk
- Make smarter HR decisions

Core Message

- Organizations that proactively support employee wellbeing are more likely to retain talent, sustain productivity, and build resilient teams.



Thank You

ANY QUESTIONS?